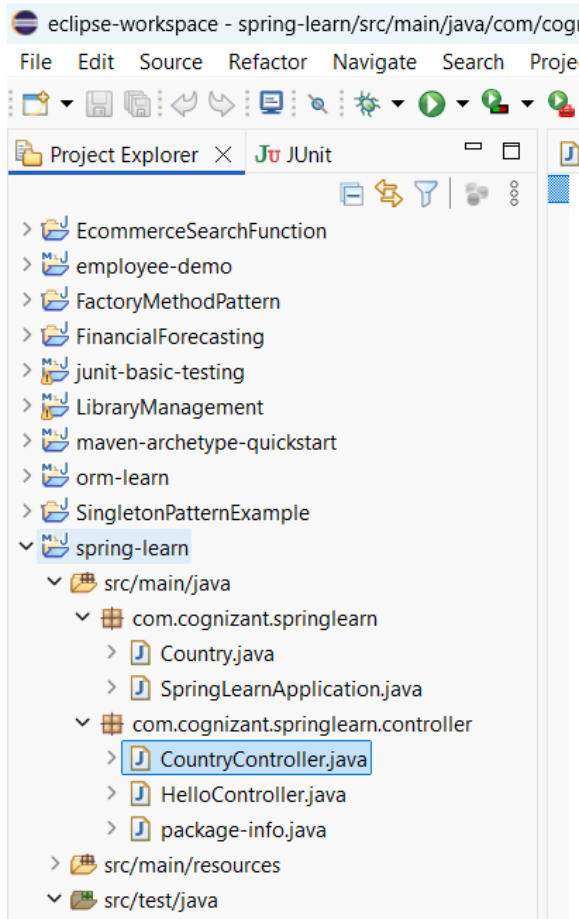


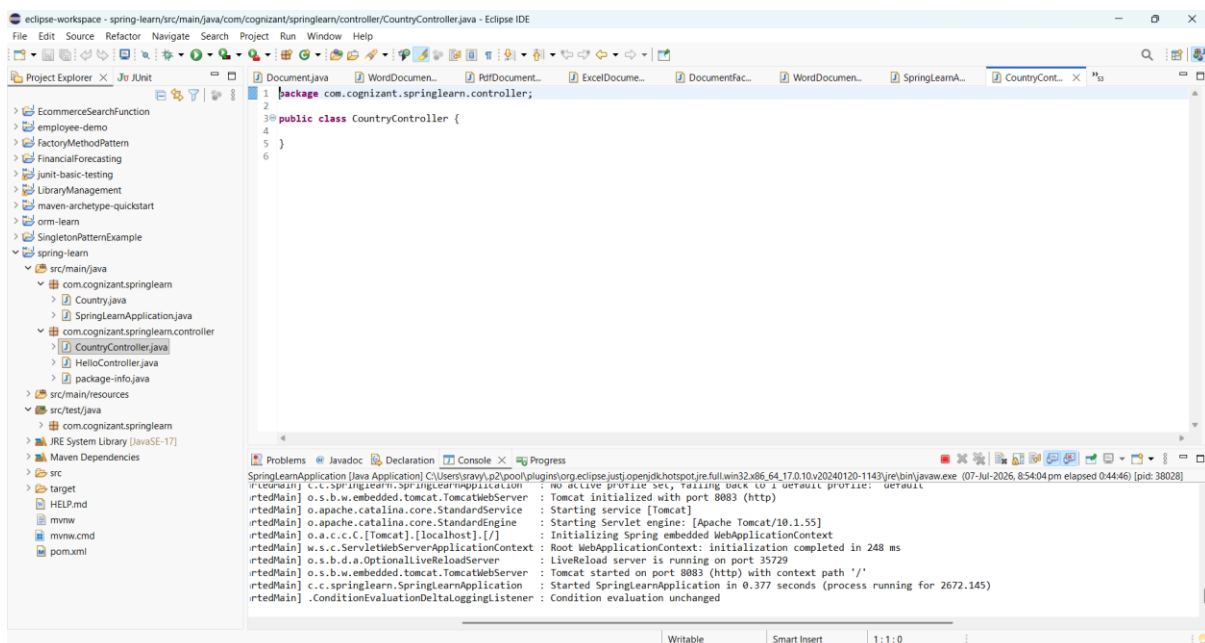
Spring REST using Spring Boot 3

REST - Country Web Service

Step 1: Project Structure – Spring REST Application

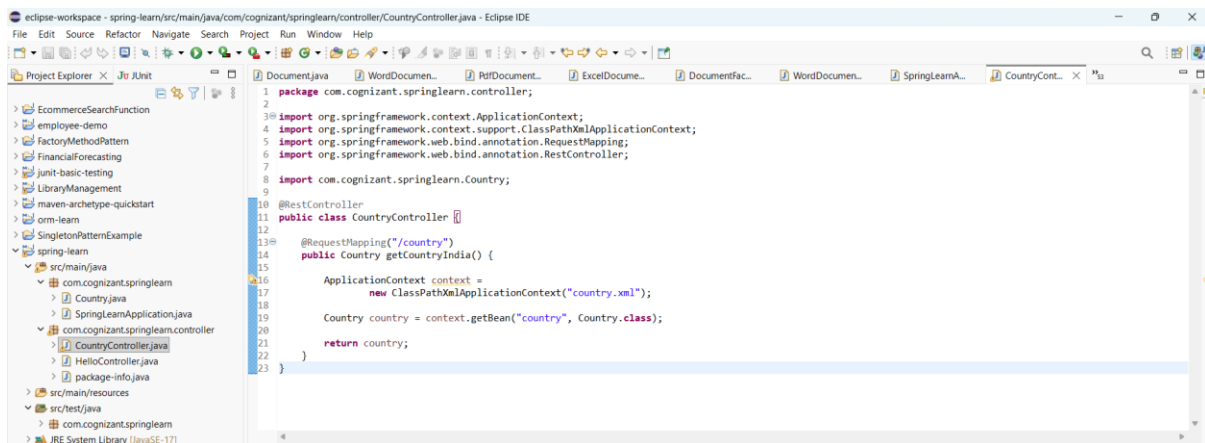


Step 2: CountryController.java – REST Controller



COGNIZANT DIGITAL NURTURE 5.0 – JAVA FSE

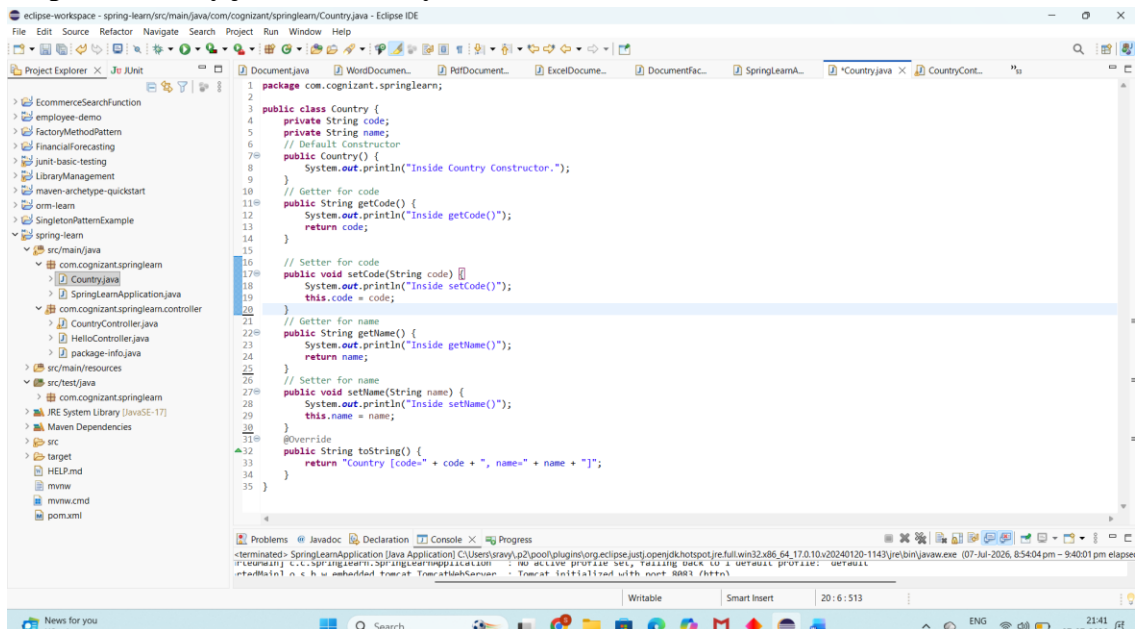
Step 3: CountryController.java



The screenshot shows the Eclipse IDE with the `CountryController.java` file open. The code is as follows:

```
1 package com.cognizant.springlearn.controller;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5 import org.springframework.web.bind.annotation.RequestMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 import com.cognizant.springlearn.Country;
9
10 @RestController
11 public class CountryController {
12
13     @RequestMapping("/country")
14     public Country getCountryIndia() {
15
16         ApplicationContext context =
17             new ClassPathXmlApplicationContext("country.xml");
18
19         Country country = context.getBean("country", Country.class);
20
21         return country;
22     }
23 }
```

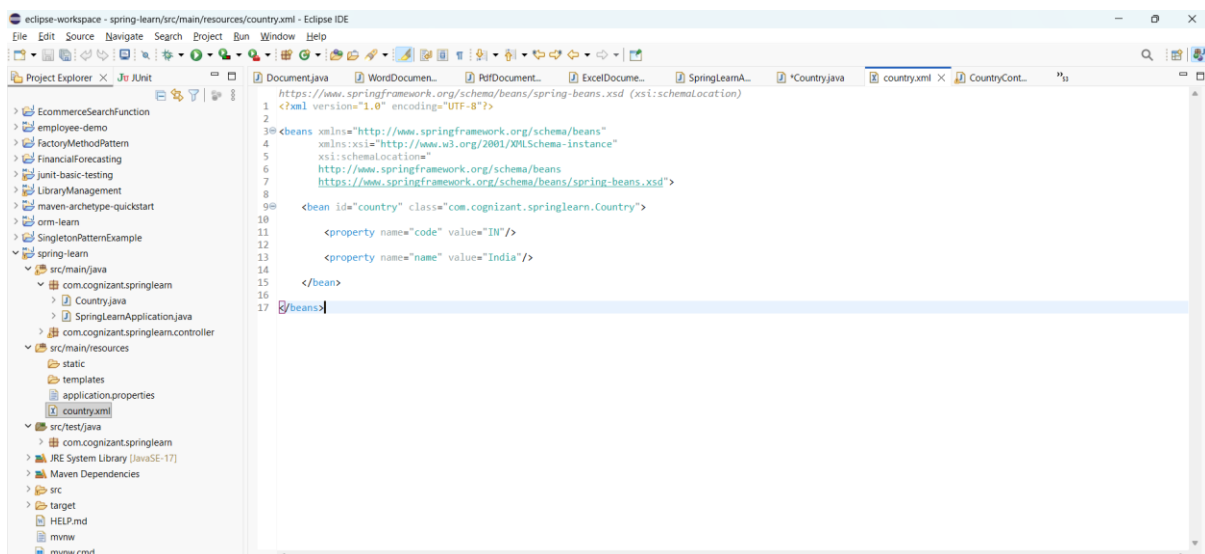
Step 4: Country.java – Country Bean Class



The screenshot shows the Eclipse IDE with the `Country.java` file open. The code is as follows:

```
1 package com.cognizant.springlearn;
2
3 public class Country {
4     private String code;
5     private String name;
6     // Default Constructor
7     public Country() {
8         System.out.println("Inside Country Constructor.");
9     }
10    // Getter for code
11    public String getCode() {
12        System.out.println("Inside getCode()");
13        return code;
14    }
15    // Setter for code
16    public void setCode(String code) {
17        System.out.println("Inside setCode()");
18        this.code = code;
19    }
20    // Getter for name
21    public String getName() {
22        System.out.println("Inside getName()");
23        return name;
24    }
25    // Setter for name
26    public void setName(String name) {
27        System.out.println("Inside setName()");
28        this.name = name;
29    }
30    @Override
31    public String toString() {
32        return "Country [code=" + code + ", name=" + name + "]";
33    }
34 }
35 }
```

Step 5: country.xml – Spring Bean Configuration

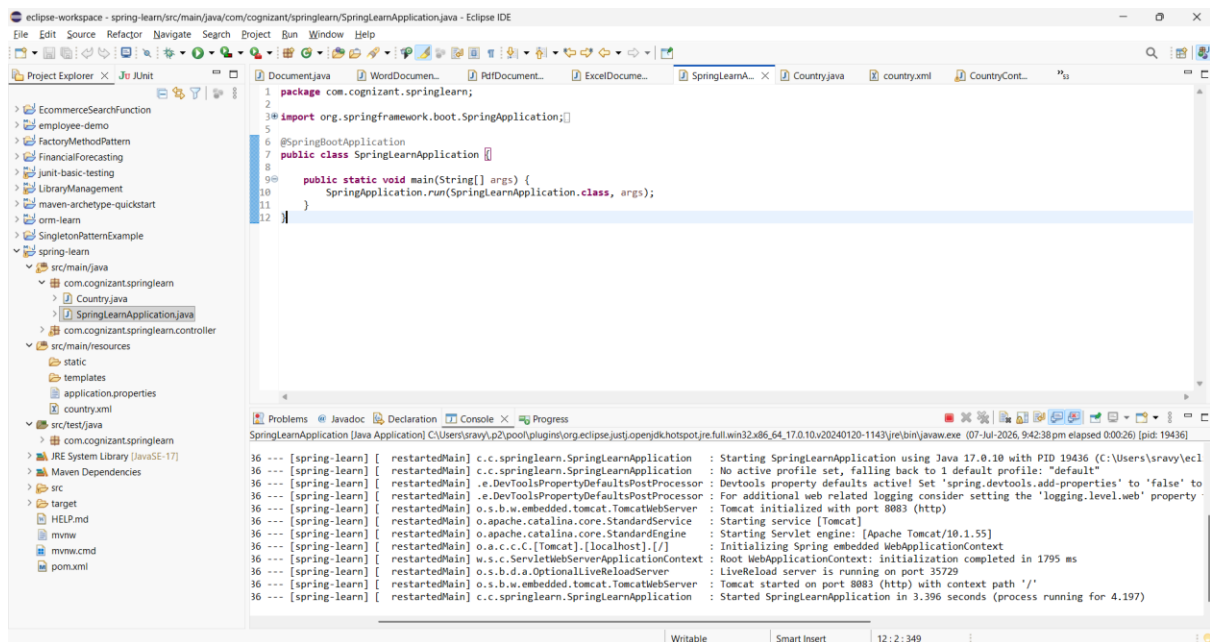


The screenshot shows the Eclipse IDE with the `country.xml` file open. The XML content is as follows:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="country" class="com.cognizant.springlearn.Country">
10
11         <property name="code" value="IN"/>
12
13         <property name="name" value="India"/>
14     </bean>
15 </beans>
```

COGNIZANT DIGITAL NURTURE 5.0 – JAVA FSE

Step 6: SpringLearnApplication.java – Main Spring Boot Application



The screenshot shows the Eclipse IDE with the SpringLearnApplication.java file open. The code is as follows:

```
1 package com.cognizant.springlearn;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class SpringLearnApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(SpringLearnApplication.class, args);
11     }
12 }
```

The console output shows the application starting successfully:

```
SpringLearnApplication [Java Application] C:\Users\srahy\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.17.0.10.v20240120-1143\jre\bin\javaw.exe (07-Jul-2026, 9:42:38 pm elapsed 0:00:26) [pid: 19436]
36 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.10 with PID 19436 (C:\Users\srahy\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.17.0.10.v20240120-1143\jre\bin\javaw.exe)
36 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
36 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to
36 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
36 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
36 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
36 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
36 --- [spring-learn] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
36 --- [spring-learn] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1795 ms
36 --- [spring-learn] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
36 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
36 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 3.396 seconds (process running for 4.197)
```

Step 7: Browser Output – REST Country Web Service Response



The screenshot shows a web browser with the URL `localhost:8083/country`. The response is a JSON object:

```
{ "code": "IN", "name": "India" }
```

COGNIZANT DIGITAL NURTURE 5.0 – JAVA FSE

Step 8: Postman Output – REST Country Web Service Response

The screenshot displays the Postman application interface. The main workspace shows a GET request to `http://localhost:8083/country` that has been executed successfully, returning a `200 OK` status. The response body is a JSON object with the following structure:

```
{  "code": "IN",  "name": "India"}
```

The interface includes a left sidebar with 'COLLECTIONS' and 'ENVIRONMENTS' sections. The bottom status bar indicates the system time as 23:00 on 07-07-2026.